Python OOP Practice Questions

Question 1: Food Delivery Order System

Build a food delivery order system using Python OOP.

Classes to Create

- Customer
- FoodItem
- FoodOrder
- OrderFileManager

Rules

- Customer has wallet balance.
- FoodItem has price and available quantity.
- FoodOrder should check quantity, food availability, and wallet balance before placing the order.
- If the order succeeds, wallet balance should reduce and food quantity should reduce.
- If the order fails, proper order status should be stored.
- OrderFileManager should save every order into **food_orders.txt**.

Test Cases

- 1 successful order.
- 1 failed order due to low wallet balance.
- 1 failed order due to unavailable food quantity.
- 1 failed order due to invalid text quantity.
- 1 failed order due to zero quantity.

Expected Saved File Content

- Customer Name
- Food Item Name
- Quantity
- Total Amount
- Order Status

Question 2: Movie Ticket Booking System

Build a movie ticket booking system using Python OOP.

Classes to Create

- Customer
- Movie
- Booking
- BookingFileManager

Rules

- Customer has wallet balance.
- Movie has ticket price and available seats.
- Booking should check seat count, seat availability, and wallet balance before confirming the booking.
- If the booking succeeds, wallet balance should reduce and available seats should reduce.
- If the booking fails, proper booking status should be stored.
- BookingFileManager should save every booking into **bookings.txt**.

Test Cases

- 1 successful booking.
- 1 failed booking due to low wallet balance.
- 1 failed booking due to unavailable seats.
- 1 failed booking due to text input for seats.
- 1 failed booking due to zero seats.

Expected Saved File Content

- Customer Name
- Movie Name
- Seats Booked
- Total Amount
- Booking Status

Submission Requirements

- Python code file or Google Colab notebook.
- Screenshot of output for all test cases.
- Screenshot or copy of the saved text file content.
- Short explanation of the classes created and how they are connected.